

Technical Report of 23rd Indian Scientific Expedition to Antarctica—An overview

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The Technical Report of 23rd Indian Scientific Expedition to Antarctica (IASE) for the year 2003 – 2004 is poised at a poignant juncture of Indian Scientific Expeditions to Antarctica. As Expedition Leader and Station Commander of 23rd IASE, it is indeed a good fortune to witness a paradigm shift from a change at the helm and a complete reorientation of the scientific programmes resulting in a major quality jump and value addition. As this report is being prepared and submitted, it is appropriate to highlight these changes and value additions put in place by the National Center for Antarctic Research and Ocean Studies – NCAOR – the mother of / umbrella of all scientific programmes in Antarctica – and now, the Arctic too.

NCAOR is currently going through some overhauls in its decade long / 10-year long existence, to adopt to the new requirements of Polar science. This also coincides with the coming into existence the Ministry of Earth Sciences, Government of India. This also heralds the significant scientific feat of leaping, as it were, from the Antarctic to the Arctic. A great deal of reorganization has already taken place in the last three years / two years across scientific programs, state-of-art Laboratory facilities and a strategic new Logistics Programme to meet new requirements and challenges. The emphasis on collaboration with young scientists / research workers for advancing the frontiers of Polar Science is only one of the several such paradigm changes taking place at this point of time.

Climate change and key related issues of Biodiversity and sustainability are by far the biggest issues the world is facing today. There is an unprecedented surge of international awareness about climate

change, energy, oil, food production, biodiversity and degradation of natural resources. NCAOR has clearly focused on these burning issues as there is a new urgency and stakes have become higher.

REPORT OF XXIII INDIAN ANTARCTIC EXPEDITION

The present report is a collection of scientific papers that have emerged from the scientific programs carried out during the austral summer of year 2004 (January to March) followed by the polar winter of 2004 (April to December). In essence, the report pertains to work in continuation of the on-going programmes of the previous year, with some new additional programmes.

The scientific papers have been grouped under (a) Meteorology and Atmospheric Sciences; (b) Earth Sciences, Glaciology & Global Change; (c) Biology and Environmental Sciences; (d) Engineering and Communication; (e) Human Physiology and Medicine; and (f) Logistics Support. The multi-disciplinary nature of scientific programmes with their core participating organizations and supporting institutions are shown in the following table.

(a)	Earth Sciences, Glaciology & Global Change	NGRI, GSI, SOI, NHO
(b)	Meteorology and Atmospheric Sciences	IMD, NPL, IIG
(c)	Biology and Environmental Sciences	BSI, NCAOR
(d)	Engineering and Communication	DEAL, EIL, SERC
(e)	Human Physiology and Medicine	DIPAS, CCRAS
(f)	Logistics Support	Indian Army (EME, ES.3); MI Room

SIGNIFICANT HIGHLIGHTS

Without falling into the trap of “first time” metaphor, there have been some new programmes with a freshness of approach and a historic visit by the Hon’ble Minister of Science and Technology to Antarctica.

(a) Earth Sciences, Glaciology & Global Change

The first ever reference gravity station at Indian Antarctic base in Maitri, was successfully established by National Geophysical Research

Institute (NGRI) Precise Absolute Gravity measurements for deformation studies were obtained. The most crucial requirement of maintaining a temperature within the range of 158 to 258 Celsius was achieved by installing auto-cut-off heater during Absolute Gravity measurements. We have also connected earlier gravity observations with AGM reference point and reoccupied five GPS stations of Survey of India by digital relative Gravimeter. Continuous magnetic observations were also made for 20 days and a magnetic traverse across litho units in Schirmacher area carried out.

A second team of NGRI carried out the on-going programme of work pertaining to: (i) Daily operation of the Seismological Observatory, (ii) Daily operation of the GPS Tracking Station at MAITRI, (iii) Storage, Preliminary Analysis of Data periodically, and (iv) Backup of the Digital Seismic Data and GPS Data in addition to regular checking of the Systems, UPS, Batteries and Power supplies, etc.

The National Hydrographic Office (NHO) collected bathymetric and physical oceanographic data (sound velocity, sea temperature profile and water conductivity) on board MV Emerald Sea using Acoustic Current Meter and CTD probe. A total of 250 nautical miles of oceanographic survey was completed. Geodetic and meteorological observations were also undertaken during this period. A total of 22 CTD probe dips - 15 in ocean and 7 in lakes - both at India Bay, Priyadarshini Lake in Maitri and during task force voyage to Vestford and Larsemann Hill areas was carried out to measure salinity depth and conductivity of lakes.

Survey of India (SOI) identified fourteen sites in the planned area of Schirmacher Oasis. Concrete pillars of one cubic foot with brass plug were erected as a monument in all the places. GPS observations were made at all the points with respect to the base station Maitri-S. GPS data will be analyzed in the head quarters at Dehra Dun.

The Geological Survey of India (GSI) team carried out studies pertaining to geological, glaciological and associated work in Gruber mountains and Schirmacher Oasis.

(a) Geological mapping of Gruber anorthosite complex revealed an outer rim of plagioclase megacryst-bearing noritic anorthosite to norite and a core occupied by medium to coarse-grained leucocratic anorthosite with accessory orthopyroxene and ilmenite. The rim portion shows rare metric-scale ilmenite-rich layers. The megacrystic noritic anorthosite is usually

foliated, trending broadly E-W with steep southerly dips in the northern rim area of the complex. Foliation is defined by stretched and aligned plagioclase megacrysts with mafics and feldspar, forming anastomosing fabric around the megacrysts. The western contact of the anorthosite with the N-S trending foliated charnockitic rocks was observed in an area close to the Peterman range. From the presence of rafts and enclaves of megacrystic noritic anorthosite within the leucocratic anorthosite, the latter has been inferred to intrude the megacrystic noritic anorthosite. A 500 m long basaltic dyke and several lamprophyre dykes were mapped and sampled in the western part of the Gruber mountains. Several discrete mylonites, both layer-parallel and also trending at high angles to the foliation in the noritic anorthosite were observed. The rocks are intensely jointed in two major directions, broadly trending N-S and E-W with subvertical dips. Subhorizontal sheet joints were also observed at several places. Selective weathering in the glaciated terrain along joints has accentuated the alpine topography resulting in a jagged and highly rugged relief of the Gruber Mountains. During mapping 62 rock samples were collected for petrological, geochemical and geochronological studies.

(b) Geological work in Schirmacher Oasis centered mainly around the post-tectonic lamprophyre and basaltic dykes. The lamprophyre dykes include biotite and amphibole-phenocryst bearing aphinitic rocks and alkaline lamprophyres. The biotite phenocryst-bearing lamprophyres have an E-W trend whereas the basaltic dykes generally trend N-S. A total of 40 samples were collected to study the petrological, geochemical, geochronological and palaeomagnetic aspects. This integrated approach will be useful in determining palaeolatitudes for regional correlation. A notable feature has been the occurrence of white powdery encrustations of 'marbilite' on various rocks. Eight samples have been collected to study its geochemistry that may indicate role of wind-blown sea water evaporated salts *vis-à-vis* salinity of lake water in Schirmacher area.

(c) Glaciological work: Iceberg monitoring was continued onboard the ship MV Emerald Sea during the onward journey in December 2003. The first iceberg was sighted at Latitude 44° 47' S and Longitude 178° 51' E. A total of 193 icebergs of various shapes (weathered, tilted and tabular) were recorded between latitudes 44° 47' and 69° 29' S. The polar ice front - monitoring was continued during the present austral summer in the DG snout area and in its western continuity using earlier reference points. Out

of 13 points, 11 recorded an annual retreat (base Feb. 2003) ranging from 0.38 m to 14 m. At two locations an advance of 0.57 m and 5.27 m was observed. In the DG snout area 15 points were monitored which revealed annual average retreat of 3.51 m with range between 0.87 m (point 13) and 10.19 m (point 5). Snow accumulation studies in Dakshin Gangotri will be monitored beginning from March.

(d) Lake sediment and Water samples for geochemical assessment were collected from proglacial, land locked and epishelf lakes in continuation of the ongoing programme. Various parameters such as temperature, pH and TDS were recorded using portable pH and TDS meters. A total of 33 lakes have been sampled. A core sample of about 50 cm was obtained from Priyadarshini Lake for sedimentological studies.

(e) GPS surveys were carried out in collaboration with Indian Institute of Geomagnetism (IIG) in the Schirmacher and adjoining *nunataks*. A total of 14 marker points on continental ice sheet and its northern front were monitored using high precision GPS measurements for deciphering movement of ice sheet. In addition 19 more GPS points in surrounding *nunataks* were recorded for assessing regional crustal deformation in this part of the Antarctic Shield.

(f) TL Dating: Four terrace sediment samples were collected from Priyadarshini Lake area for Thermoluminescence (TL) dating, representing clay-rich layers sandwiched between glacier boulder horizons at 116 m and 109 m datum levels ASL to estimate rate of deglaciation since the last glacial maxima.

(b) Meteorology and Atmospheric Sciences

The Indian Meteorological Department (IMD) carried out (a) surface observations for weather parameters like wind speed, wind direction, visibility, air temperature, atmospheric pressure, clouds by recording at every three hourly interval and six hourly observations of main synoptic hours were transmitted to IMD, New Delhi over telex, (b) daily weather summary displayed on board, (c) provided weather forecast services for fieldwork parties and helicopter sorties and meteorological data was supplied to all summer team members and their institutions for their expedition work, (d) three ozonesonde ascents were computed for vertical profile of ozone and one radiometer sonde ascent was attempted, (e) Brewer spectrophotometer developed defects due to snapping of data cable at

computer end which was rectified and made operational subsequently, (f) The installation for diffused solar radiation stand in concrete foundation along with pyranometer, shading ring assembly in observatory enclosure and the installation of its pyranograph in IMD Lab was completed and recording started on 6th February 2004, (g) Defects in wind speed recorder and temperature sensor during this month were rectified and made operational.

The National Physical Laboratories (NPL) carried out (a) Measurement of green house gases using Gas Chromatograph was set up and data has been collected on regular basis for CO₂ and CH₄. Data for Methane and CO₂ have been found to be in good agreement. (b) Sun-Photometer – MICROTOP incorporating five channels to maneuver the solar radiation at 300, 305, 312, 940 and 1020 nm was set up to estimate the ozone on all clear sunny days on hourly basis. The first three channels were adapted for ozone computation while other two for water vapor and aerosol optical depth. (c) Technical problems in the CO Analyzer system were rectified and data collection resumed as part of winter programme. Results: (i) Surface concentration of Atmospheric Methane (CH₄) was found to be in the range of 1.69 to 1.72 ppm. (ii) Water vapor concentration was found to be in the range of 0.3 to 1.2 cm. (iii) Monthly mean total column ozone was found to be around 298 DU.

The Indian Institute of Geomagnetism (IIG) carried out (a) Proton Precession Magnetometer (PPM) recording of Total Magnetic Field intensity (F) has indicated a drop of about 120 nT per year (as of 2002), as compared to global average drop of 30 nT per year. Our earlier study of high-latitude stations in the northern and southern hemispheres has shown that this decline is confined to a narrow horseshoe shaped belt in Antarctica, encompassing Maitri. The rate of decline falls in the poleward and equatorward directions. (b) A Digital Fluxgate Magnetometer (DFM) was proposed to be installed as unmanned Magnetic observatory in Wohlthat mountains but in the absence of a *Portacabin*, at this point of time, this is being installed at Dakshin Gangotri and operated throughout the winter. A chain of unmanned magnetic observatories is being attempted to study and monitor the movement of small-scale auroral current systems. (c) Riometer and Magnetometer data has been collected in analog and digital mode on a continuous basis. (d) 19 existing GPS stations setup during 2003 were reoccupied during this summer and data collected from each station for a period of 72 hours to study crustal movement. New markers were fixed for

21 GPS stations setup on the polar glacier and data collected for 48 hours at 15 locations. One GPS was operated permanently at Maitri to provide the baseline data. The data will be used for TEC, scintillation and water vapor content and other atmospheric studies.

(c) Biology and Environmental Sciences

Botanical Survey of India (BSI) carried out field-work during the XXIII Indian Scientific Expedition to Antarctica in the year 2003 – 2004. Algae growths are abundant and readily visible on the surface of the rocks, boulders, weathered soils and moss carpets. The author collected pieces of moss from which upper layer of bottom sediments were removed using spoon and water samples from different lakes of Schirmacher Oasis, East Antarctica to study the diatom flora. The collected samples was fixed with 4% Formalin solution and brought to Botanical Survey of India, Port Blair for further study. Part of the samples was oxidized in a mixture of sulphuric acid and potassium dichromate (volume 3:1) to remove the organic matter. Then the material was centrifuged and digested again. It was then rinsed with distilled water several times, to get rid of acid. The cleaned samples were examined under light microscope (Olympus CX 41) and Taxa were identified using recent references All collected samples have been deposited in Botanical Survey of India, Port Blair for further reference.

The development of micro algae is generally triggered by light, temperature and nutrients. During the study period physicochemical properties were recorded from the sampled areas and the data are given in the Table 2. The atmospheric temperature was -4.5°C to +10.5°C and the water temperature was -1°C to + 9.5 °C. The lake water salinity ranging from 0.25 to 24.2ppt (this 24.2ppt recorded in land locked lakes only) and the pH value from 6.1 to 9.5. Due to higher salt concentration the algal diversity is more in land lock lakes compared to the proglacial and periglacial lakes. The current study revealed that the cosmopolitan species i.e *Pinnularia borealis*, *Hantzschia amphioxys*, *Nitzschia obtuse* and *Luticola muticopsis* are presented in this Oasis but distribution of diatoms are differing from varies ecological factors. The western part of the land lock lakes harboring more variety of diatoms than the other part of the Oasis.

The National Institute of Antarctic and Ocean Research (NCAOR) succeeded in Lake sediment coring for the first time in Antarctica. The ecological study shows that *Cosmarium* alga was dominant in composition in three sediment cores. Pigment profiles of sediment cores from top, central,

and bottom part are the same. The top level of the core has relatively less pigment in quantity than the central part of the core, while the bottom level of core has even lesser quantity. The large pigment system assimilates at central level than the top level due to leaching, and more habitation microbial population in central (middle) part of the core. At bottom level due to degradation of pigment over a longer period of time the quantity of pigment is the lowest. Mineral matter, Mn and Fe concentration do not show any correlation from top to bottom.

A preliminary microscope analysis showed that *Cosmarium* spp. (green algae from group of *Desmidiaceae*) predominated (72%) in all three-core sediment samples. Other microbial remnant diversity comprised of algal cysts (*Zygnema*) and fungal spores (Helminthosporite sp. *Alternaria*) abundance in sediment layers. The organic contents of the core are mainly of algal cysts. Fungal spores probably derived from air got trapped during sedimentation. Mn, Fe metal concentration is not showing any correlation from top to bottom.

(d) Engineering and Communication

Engineers India Limited (EIL) carried out work related to Assessment of Structural Strength of Maitri Building. Collection of all available data on Maitri station structure was followed by validation and compilation of collected data and gathered information. Carried out and completed (a) Detailed visual inspection of various structural components of the stilt of Maitri building, (b) Physical measurement of structural components of the stilt and superstructure, (c) Grid marking and preparation of working sketches in AutoCAD, (d) Visual inspection of the composition of wall, floor and roof panels of Main Block, A Block, B Block and C Block (e) Inspection of base connection under columns by removal of the back-filled earth up to founding level, (f) Study of damages found in structural columns along grid by removal of the back-filled earth below existing ground level, and (g) Detailed inspection of floor panel in the main block passage area after removal of the false flooring.

Another item on 'Review of Liquid Effluent Management System at Maitri' was also taken up by EIL and completed as per schedule of objectives. A format for data requirement was devised and handed over from EIA laboratory.

Environmental monitoring was carried out by NCAOR for air and water component during the austral summer of 2003-2004 at the Indian

Polar Research Station “Maitri” in compliance with the statutory requirements of the Protocol on Environmental Protection to Antarctica Treaty. The objectives zeroed in on air monitoring, water and wastewater sample analysis and efficacy assessment of the Rotating Biological Contactors (RBC) and enhancing the EIA laboratory and relocating it to a suitable place inside the Maitri station.

During summer of the XXIII expedition, Defence Electronics Applications Laboratory, Dehradun (DEAL) successfully commissioned Nera Saturn M INMARSAT Terminal installed for new e-mail connectivity with Internet accessing facility and established voice and data communication link using CODAN HF Adaptive Transceiver.

(e) Human Physiology and Medicine

Central Council for Research on Siddha and Ayurveda (CCRAS) carried out a new programme of work on “The Effect of certain established Ayurvedic drugs (*Rasayana*) as food supplements on aging process in cold climatic conditions of Antarctica” as per statement of objectives. For this project, 21 subjects among the team members of 23rd IAE were registered out of which 13 continued the regimen for the stipulated period of 6 weeks. Various parameters like lean body mass index, MAC (mid arm circumference), Blood pressure, Pulse rate etc. were taken before starting the study and repeated at an interval of every fortnight. Parameters under visual analog scale (VAS) like dizzy spells / giddiness, breathlessness on exertion, constipation, urgency to micturate, aching muscles and joints, lumbago or shoulder pain, numbness, tremors, sleep abnormalities, loss of appetite and irritability were also observed. Besides this the blood samples were collected and analyzed for routine pathology and chemistry of the serum to evaluate the effect of the regimen on liver and kidney functions, enzymes, immuno-globulins and free radicals.

Defence Institute of Physiology and Medicine (DIPAS) continued their on-going programme of work. Baseline readings were obtained for various parameters namely Blood Pressure for 48 hours at every 2 hours interval, Hand Grip Strength for 48 hours at every 2 hours interval, Monitoring of Heart Rate for 24 hours, Estimation of Energy Expenditure while normal walking etc. The subjects were so selected that their data could also be obtained while returning from Antarctica, thus the selection was made from among the summer team. The second phase of experiments were repeated after spending about 20 days at the Maitri station so as to provide time for

the body to make adjustments, to suit the Antarctic environment. Estimation of Energy Expenditure was done simultaneously with measurement of Physical Fitness Index (PFI).

The MI Room for the Winter Station at Maitri was strengthened with the following additions and improvements: (i) The layout of the Medical Store in the loft area was revamped. All equipment for use in the Operation-Theatre was shifted to the A Block OT complex. Old expired medicines were removed from the store and shifted to a container for back-loading. The remaining medicines were laid out group-wise. Medicines of common use were shifted to the MI Room and displayed category-wise for ease in dispensing. (ii) Lack of filled oxygen cylinders, as detailed in the monthly report of January, was projected to NCAOR and six cylinders were received by IL-76 aircraft. (iii) A new Ultrasonography machine, which was brought along with the XXIII IAE, was set up in the Maitri hospital and was made fully functional. This equipment can prove to be a very useful diagnostic tool. However, doctors selected for the Antarctica expeditions need to be given basic training in using this equipment and interpretation of the same. (iv) A Semi Auto-Analyser, which was lying unused in the hospital for more than a year, was also made functional and is now being utilized for carrying out investigations. Doctors selected in future need to be trained in operating this equipment also. (v) There were about 20 patients with minor ailments, who have recovered completely and the morale of the team remained high during the period under report.